Wang 2019 Glioblastoma — analysis summary

RUN 2 OF 8

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Judged on the embedding + lobe composition (Kürten fork: never a scalar). Harmony on sample cut the same-sample neighbour fraction 0.99→0.6 and dissolved the θ=2 monochrome patient lobes: one connected component, no lobe a single patient (dominant share 0.35; 6–7/7 samples per lobe). Mixing is partial (iLISI median 1.9/7) — expected GBM CNV identity within a continuous multi-sample manifold, not an archipelago.

Gate 2 — module geography → GO (non-branching)

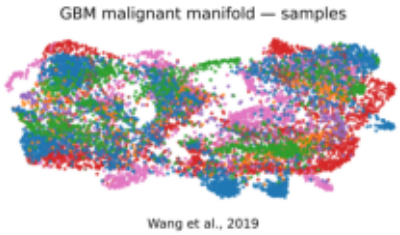
PPT tree over-branches a continuous cloud (forks 5→18 with node count, none stable), and the two arm-candidates Gast & pSom co-vary (spatial r=+0.16) — no bifurcation. Shared Gast/pSom gradient with neural crest as the inverse terminal end (Crest–pSom r=−0.87; Crest-high decile at pseudotime 0.84). Non-branching pattern present.

Gate 3 — hub cross-repression → NA

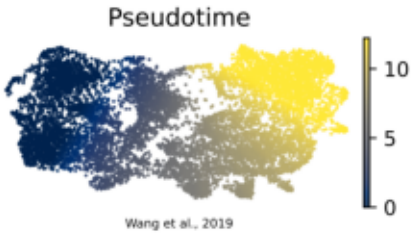
Not applicable — the manifold is non-branching (single principal curve, 0 forks), so there is no hub bifurcation to test Gast↔pSom cross-repression.

STEP FIGURES

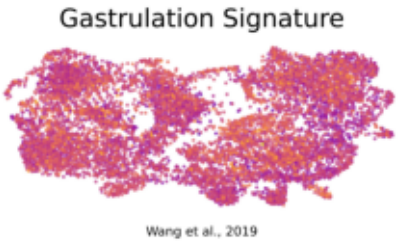
Manifold — samples



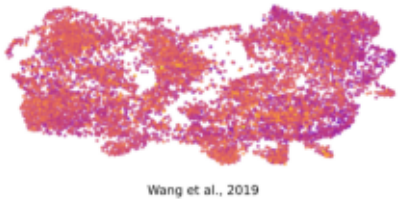
Pseudotime



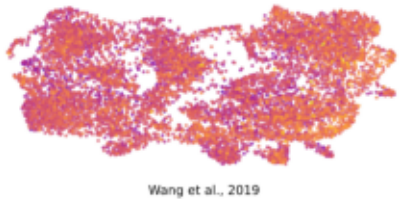
Gastrulation



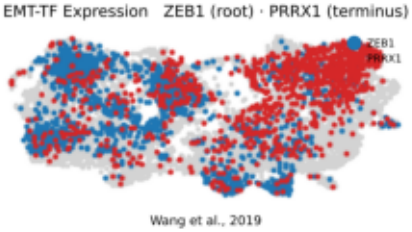
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
IDH-wildtype glioblastoma (Wang et al., 2019), 10x — 9,634 malignant cells (7 samples); non-branching developmental axis.

Verdict.
Pattern reproduces (non-branching): one continuous gastrulation/somitogenesis → neural-crest developmental axis, EMT-TFs ZEB1 → PRRX1. Gate 1 GO / Gate 2 GO / Gate 3 NA.

Parameter choices

Threshold	min 237 cells/sample (largest multiplicative gap 396→143, 2.8×) → 7/10 samples, 9,650 cells	**Integration**	Harmony θ=10 on sample; n_pcs=20; UMAP spread=1.3; overdispersed=2,113
Islands	16 detached cells removed (radius graph) → 9,634 on one component	**Geometry**	Linear-pSom/Crest non-branching curve (ElPiGraph N=20, λ=0.3, μ=100); fold-free (SA 2.8); PPT 0 stable forks
Root	curve tip 2 = Gast/pSom-high, Crest-low origin (neural crest late terminus)	**Scope**	whole manifold; cell cycle orthogonal (|corr|<0.1)
EMT-TFs	ZEB1 (root, 70%) → PRRX1 (terminus, 40%); Snail/Twist/FOXC <5%		